[PLACEHOLDER: Title TBD]

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Housing and Fertility

[PLACEHOLDER: motivation pitch — housing access shapes fertility]

- [PLACEHOLDER: motivation fact 1]
- [PLACEHOLDER: motivation fact 2]
- [PLACEHOLDER: motivation fact 3]

This Paper

[PLACEHOLDER: this paper]

- [PLACEHOLDER: theory contribution]
- [PLACEHOLDER: quantitative contribution]
- [PLACEHOLDER: policy contribution]

Related Literature

- [PLACEHOLDER: Sommer–Sullivan–Verbrugge slot]
- [PLACEHOLDER: Doepke–Kindermann slot]
- [PLACEHOLDER: Greaney–Parkhomenko–Van Nieuwerburgh slot]
- [PLACEHOLDER: Couillard slot; Hacamo slot]

A Simple Model: Demographics and Timing

A stationary two-period overlapping-generations economy. A potential young household enters with a type

$$i = (y_i, a_i),$$

disposable income y_i and entry wealth a_i inherited from the old who exit as she arrives.

- Young households choose whether to enter the market, then tenure, occupied space, number of children, and saving.
- A young owner carries her house into old age and becomes next period's old incumbent; a young renter reaches old age with no house.
- Old households choose consumption, how much housing to keep, and the estate they leave, then exit.

Entry is a discrete choice against an outside value $\bar{W}^E$ with taste shocks of scale κ_E, so who reaches owner housing is set at entry. Births plus a renewal inflow form the next young cohort, and the exiting estates fund its entry wealth.

Preferences

Children compete with their parents for the same budget and the same house: each child claims χ units of consumption and κ units of space. Over consumption c_i, occupied housing h_i, and number of children n_i, the young household values

$$u_i^Y(c_i, h_i, n_i) = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i, \qquad \alpha, \vartheta > 0.$$

One more child means less space and less consumption left for the adults.
Old households value consumption, retained housing, and a warm-glow estate e_j,

$$u^O(c_j^O, h_j^O, e_j) = \log c_j^O + \gamma \log h_j^O + \mathcal{B}(e_j),$$

and a young household discounts her own old age by β. Saving carries $a_i' \geq 0$ into old age at gross return $1 + r$; the savings load $k \equiv \beta(1 + \gamma + b)$ is old age's claim on each unit of adult consumption.

Markets and Technology

One fixed stock $\bar{H}$ of divisible floorspace, occupied by renting (R) or owning (O). The segments differ only in the largest size available,

$$\mathcal{H}^R = \{0 < h \le \bar{h}^R\}, \qquad \mathcal{H}^O = \{0 < h \le \bar{h}^O\}, \qquad \bar{h}^R < \bar{h}^O,$$

so family-sized space must be bought, not rented. All space clears in one market, and a no-arbitrage user cost ties the rental rate to the price:

$$q = \rho P, \qquad \rho = r + \delta + \tau^p.$$

The renter pays q as rent; the owner pays it as interest, tax, depreciation, and forgone return — equal per-unit value by arbitrage. Selling a house triggers a capital-gains tax that death forgives through basis step-up, a per-unit wedge on housing sold while alive,

$$\bar{\ell} = \tau^g \, \frac{P\,\varpi}{1+\varpi},$$

the only place the model touches inflation.

The Household Problem, Mode by Mode

Conditional on entry, the household picks a tenure mode $m \in \{R, O\}$ and solves

$$W_i^m = \max_{c_i, h_i, n_i, a_i'} \; u_i^Y(c_i, h_i, n_i) + \beta\big[\mathbf{1}\{m = R\} V^R(a_{j'}) + \mathbf{1}\{m = O\} V^O(a_{j'}, h_i)\big]$$

subject to one budget and, for buyers, two financing tests:

$$c_i + qh_i + a_i' = y_i + a_i, \qquad \mathbf{1}\{m = O\}\,(1-\phi)Ph_i \leq a_i, \qquad a_i' \geq 0.$$

Space costs q per unit either way, so tenure bites only through the size limit and the down payment $(1-\phi)Ph_i$, which wealth in hand must cover. An old incumbent nets $q - \bar{\ell}$ on each unit she sells, valuing kept space *below* market by the tax she avoids; an old renter's demand is wedge-free.

Stationary Competitive Equilibrium

A stationary equilibrium is a user cost q, an asset price $P = q/\rho$, young and old allocations, entry probabilities, and stationary measures such that:

- young households solve their mode problems and choose tenure $o_i = \arg\max\{W_i^R, W_i^O\}$; old households and entry are optimal;
- the single floorspace market clears,

$$H^D(q) = \bar{H};$$

- the old-incumbent measure, the young cohort, and the cohort mass reproduce themselves, and fiscal and housing-income accounts close by lump-sum transfers.

One price equates total demand to the fixed stock. Nothing in that condition asks whether the marginal family-sized unit is held by the household that values it most — that question is the efficiency result below.

All Frictions Collapse Into One Cap

Using $P = q/\rho$, owner housing is bounded by the tighter of the size limit and down-payment capacity:

$$h_i \leq H_i^O(q, \tau^p) = \min\left\{\bar{h}^O,\ \frac{a_i \rho}{(1-\phi)q}\right\}, \qquad H_i^R = \bar{h}^R.$$

Renting, the owner size limit, and finance all reduce to one effective family-housing cap. Substituting it into the fertility first-order condition prices the marginal child:

$$\underbrace{\frac{\vartheta\left(c_i^m - \chi n_i^m\right)}{n_i^m}}_{\text{willingness to pay for a child}} = \underbrace{\chi + \kappa(q^m + \zeta_i^m)}_{\text{full price of a child}}.$$

A binding cap acts as a per-child tax of $\kappa\zeta_i^m$: children cost more exactly where family-sized space is hard to reach, with market prices unchanged.

The Access Gap

The access gap ζ_i^m is the goods value of one more unit of constrained family space, read off the household's own bundle. For a renter at the size limit,

$$\frac{u_{h,i}^Y}{u_{c,i}^Y} = \frac{\alpha\,(c_i^R - \chi n_i^R)}{h_i^R - \kappa n_i^R} = q + \zeta_i^R,$$

so she values space above the rent it commands. For an owner the gap separates finance from the size limit,

$$\zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i^O}(1-\phi)P}_{\zeta_i^{O,F},\ \text{down-payment margin}} + \underbrace{\frac{\theta_i^O}{\lambda_i^O}}_{\text{size-limit margin}} \geq 0.$$

Either way ζ_i is computable from observed consumption and housing, net of child needs: the young gap from bundles, the old wedge from deeds and the tax schedule.

Where the Cap Binds, Fertility Responds

Under equal scaling at lifetime prices, $\chi = (1+k)\kappa q^m/\alpha$, the fertility response to relaxing the cap is proportional to the same gap that measures access:

$$\Psi_i^m = \frac{\kappa}{\alpha}\,\frac{\zeta_i^m}{c_i^m - \chi n_i^m}\,\Omega_i^m, \qquad \Omega_i^m = \frac{\dfrac{\alpha}{h_i^m - \kappa n_i^m} + \dfrac{q^m}{c_i^m - \chi n_i^m}}{\dfrac{\vartheta}{(n_i^m)^2} + \dfrac{\chi^2}{(1+k)(c_i^m - \chi n_i^m)^2} + \dfrac{\alpha\kappa^2}{(h_i^m - \kappa n_i^m)^2}} > 0.$$

Holding exposure and pass-through fixed, a larger normalized gap means a larger birth response. The households closest to the cap are the ones whose fertility moves.

Why the Stock Is Misallocated

Move one unit from an old incumbent to a financially constrained young owner:

$$\underbrace{\left(q^{O} + \zeta_i^{O,F}\right)}_{\text{buyer's marginal value}} - \underbrace{(q - \bar{\ell})}_{\text{incumbent's marginal value}} = \zeta_i^{O,F} + \frac{\bar{\ell}}{1+r} + \bar{\ell} > 0.$$

The market price cancels. What is left is always positive: the young buyer's financing gap, plus her own anticipated capital-gains tax, plus the incumbent's realized one. The stock clears and is still misallocated — not because the old occupy housing, but because their tax lock-in sits next to young households with unmet access gaps. The tax wedges are transfers, not resources, so the reallocation is a Pareto improvement.

From Two Periods to a Lifecycle

The quantitative model keeps the mechanism and adds a lifecycle. A household enters at age 18 as a renter and lives 17 four-year periods to terminal age 82.

- **Income:** a five-state persistent process z, with age-dependent earnings.
- **Children:** up to two children per household, in a number-of-children and child-stage block (n, s); after a birth, children age stochastically through home stages and eventually leave, at which point their space and goods costs switch off.
- **Fertility:** completed fertility is chosen once, inside a fertile window (ages 18 to 42), subject to taste shocks — one shot, not a sequential birth hazard.

So the two-period child cost (χ, κ) becomes a lifecycle of family-size-dependent needs that arrive with a birth and expire when children mature.

Housing in the Quantitative Model

Tenure segmentation is the same object, made discrete:

$$h^R \in \{0, H_1^R, \dots\} \text{ capped at } \bar{h}^R = 6 \text{ rooms,} \qquad h \in \{2, 4, 6, 8, 10\} \text{ rooms (owner ladder).}$$

- The renter cap $\bar{h}^R = 6$ is not estimated: it is the 90th percentile of rental unit sizes, following Greaney, Parkhomenko, and Van Nieuwerburgh (2025), measured in our own ACS sample.
- Buying rung H_k at price p requires a down payment $(1 - \phi)pH_k$ with financed share $\phi = 0.80$ — the same $(1 - \phi)$ friction as the theory.
- Owners transact across rungs and pay a transaction cost; a single stock clears at one price p in general equilibrium.

Family-sized rooms (≥ 6) are still reachable mainly by owning, and still gated by the down payment.

The Household's Period Problem

Each period the household sees income and taste shocks, chooses births while fertile, then chooses tenure, size, and saving. Over a consumption/housing composite with family-size floors,

$$u(c, \mathbf{h}; m) = \frac{\left[(c - \bar{c}(m))^{\alpha}(\mathbf{h} - \bar{h}(m))^{1-\alpha}\right]^{1-\sigma}}{1-\sigma} + \psi\, \mathbf{1}\{\text{child at home}\}, \qquad \sigma = 2,$$

where the floors rise with the number of children m,

$$\bar{c}(m) = \bar{c}_0 + \bar{c}_n\, m, \qquad \bar{h}(m) = \bar{h}_0 + \underbrace{\bar{h}_{\text{jump}} \mathbf{1}\{m \geq 1\}}_{\text{first-child housing margin}} + \bar{h}_n\, m,$$

and owners enjoy a service premium $\mathbf{h} = \chi_o h$, $\chi_o > 1$, over renters' $\mathbf{h} = h^R$ (the subscript keeps this distinct from the theory section's child consumption cost χ).

- Resources are income $y(z, a)$ plus wealth; owners pay a flow user cost $(\delta + \tau^p)p$ and a transaction cost on rung changes.
- Buying is gated by the collateral bound $b' \geq -\phi\, p\, h'$, the quantitative down payment.

The floors $\bar{c}(m), \bar{h}(m)$ are the quantitative counterpart of the child costs χ, κ: a birth raises required goods and, through $\bar{h}_{\text{jump}}$, required rooms.

Equilibrium and Computation

A stationary recursive competitive equilibrium is value and policy functions, a stationary distribution μ over states (b, h, z, n, s, a), and a price p such that households optimize, μ is invariant under the induced transitions, and the single housing market clears:

$$\int h \, \mathrm{d}\mu \; = \; \bar{H}.$$

The equilibrium is solved by backward induction on the lifecycle value functions and a fixed point in the market-clearing price, with entry wealth carried in from the data.[1]

[1] The value functions are solved on a wealth grid by backward induction over the 17 periods; the stationary distribution is iterated to a fixed point and the price p is found by bisection on the market residual. The reported fit is searched on a fine wealth grid and verified on a finer one, since a coarse grid tunes to numerical noise.

What Maps to What

Every theory object has a quantitative counterpart — these set up the bridge slides.

simple model	quantitative model
user cost $q = \rho P$	user-cost flow $(\delta + \tau^p)p$ at the single price p
down-payment friction $(1 - \phi)P$	collateral bound $b' \geq -\phi p h'$, $\phi = 0.80$
effective cap $\min\{\bar{h}^O,\ a_i \rho/((1-\phi)q)\}$	owner ladder $\{2, 4, 6, 8, 10\}$ vs. renter cap 6, against the down payment
child costs $\chi,\ \kappa$	consumption floor $\bar{c}(m)$ and housing floor $\bar{h}(m)$
access gap ζ_i	shadow value of the collateral/size constraint on family space
capital-gains wedge $\bar{\ell}$	transaction cost τ on owner sales

The down payment is the load-bearing bridge: it is $\zeta_i^{O,F}$ in the theory and the $\phi = 0.80$ collateral bound in the quantitative model.

Calibration

- The fit is a 14-moment SMM table against 13 estimated parameters.
- Empirical inputs: room and ownership moments from the ACS, and lifecycle wealth, tenure, and birth-housing moments from the PSID; entry wealth is the PSID young-childless-renter distribution, carried in rather than estimated.
- Externally fixed: the financed share $\phi = 0.80$ and $\sigma = 2$.
- The rental cap is not estimated. We set $\bar{h}^R = 6$ rooms from an external rule — the 90th percentile of rental unit sizes, following Greaney, Parkhomenko, and Van Nieuwerburgh (2025), measured in our own ACS sample.[2]

The precise statement is that the p90 of renter sizes falls in the 6-room bin, so the maintained cap is a measured feature of the rental stock, not a free parameter.

[2] Search at $N_b = 120$ wealth nodes and verify at $N_b = 240$; a coarser grid tunes to numerical noise, so the reported fit is checked at the finer resolution.

The Data Do Not Fight the Cap

Re-solving at nearby cap values traces out a shallow valley around the external rule:

$\bar{h}^R$	best loss	status
5.5	7.69	converged
6.0	6.31	converged
6.5	6.24	still improving
7.0	7.15	preliminary

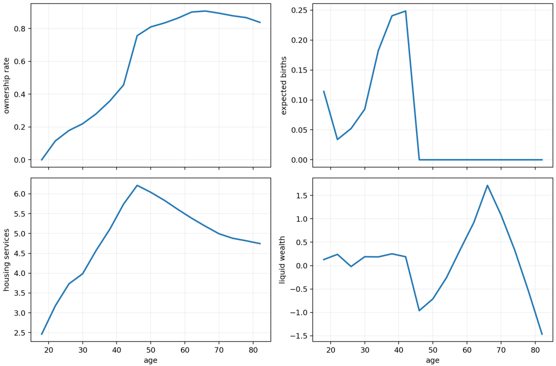
The data mildly prefer 6.5, but both flanks rise sharply, so the cap is identified and its minimum sits within half a room of the external value. We keep $\bar{h}^R = 6$ as the baseline and carry 6.5 as a robustness check.

Model Fit

moment	target	model	gap	contrib
tfr	1.918	1.947	+0.029	0.02
childless_rate	0.188	0.260	+0.072	0.10
own_rate	0.576	0.531	-0.045	0.20
own_family_gap	0.168	0.298	+0.131	0.77
housing_increment_0to1	0.664	0.617	-0.048	0.03
old parent-childless wealth gap 65–75	1.007	0.835	-0.172	0.06
childless renter mean rooms	3.805	3.656	-0.150	0.13
childless owner share rooms ≥ 6	0.596	0.595	-0.001	0.00
old nonhousing wealth/income median	2.231	2.328	+0.097	0.01
young renter wealth/income	0.179	0.354	+0.175	0.37
owner–renter mean rooms	**2.419**	**2.152**	**-0.266**	**0.85**
old_age_own_rate	**0.764**	**0.892**	**+0.128**	**2.63**
own_rate_2534	**0.341**	**0.226**	**-0.115**	**1.06**
parent 3+ vs 1–2 rooms gap	0.368	0.264	-0.104	0.09

The family-ownership moments fit well. Three misses carry the loss: old-age ownership is too high, young ownership and wealth are mistimed, and childlessness overshoots.

Lifecycle Profiles: A Deadline Purchase



- Ownership jumps at the last fertile age, not gradually before it.
- Births cluster at age 33 with a spike at the deadline; in the data they peak near 27.
- Liquid wealth drops sharply at the family purchase, as households lever up to the limit.

The remaining fit problem is timing: the model buys and gives birth too late.

What the Misses Are Telling Us

diagnosed miss	fix designed, not yet adopted
Old-age ownership is 0.892 against 0.764 in the data.	Give old owners a health, mortality, or downsizing exit.
Young ownership is 0.226 against 0.341, and young wealth is 0.354 against 0.179.	Fix fertility timing, rather than re-searching the same parameter box.
Childlessness by income runs the wrong way.	Add a child time cost, so children cost time as well as goods and rooms.

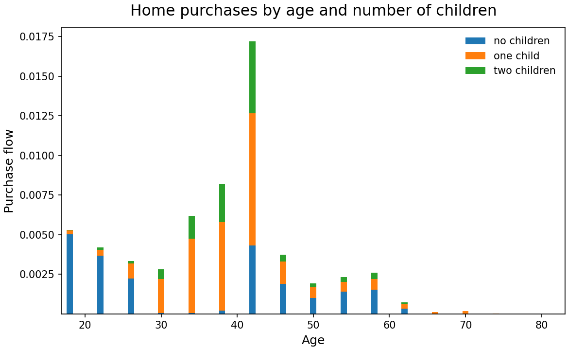
The misses hang together: late fertility and no old-age exit are doing work that no parameter can do cleanly.

The Access Gap Shows Up in the Calibration

- Among fertile renters constrained by the rental size limit, $\zeta/q = 1.28$.
- These renters value family space at $2.3\times$ market user cost.
- A share 0.734 of the fertility-response mass sits on households with $\zeta > 0$.
- The share is robust to numerical resolution, to how the targets are weighted, and to the owner ladder.

This is the equal-scaling prediction in the calibrated model: fertility exposure rises with the normalized gap $\zeta_i^m/(c_i^m - \chi n_i^m)$.

Births Trigger the Purchase at the Deadline



- About 29% of the age-42 cohort buys in that single period.
- Seven in ten of these buyers are parents, and half have a newborn.
- Buyers arrive with slack: the average one is about 10 wealth-grid cells above its down-payment threshold.

The purchase margin is a birth-timing margin, not slow saving finally crossing the constraint.

Who Holds the Family-Sized Stock

The calibrated allocation gives the empirical face of the misallocation result:

holder group	share of rooms ≥ 6 stock
post-fertile households	87%
households with no kids at home	72%
fertile parents	12.6%

The households with the largest measured access gap hold almost none of the family-sized owner stock. This is the planner-equilibrium comparison, stated as an accounting fact about the calibrated economy.

A Price Instrument Capitalizes but Barely Moves Births

- Consider a 2% tax on large homes.
- Large-home prices fall by 12.3%.
- Fertility hardly moves: $\Delta \mathrm{TFR} = +0.004$.
- The response is robust to numerical resolution, to the housing menu, and to the weighting of targets.

A pure price instrument capitalizes into house prices; it does not create births.

Birth-Linked Purchase Grants Move Fertility

A grant paid at a birth-linked purchase raises fertility, and the response grows with the size of the grant:

grant size	$\Delta\mathrm{TFR}$
0.1	+0.01
0.4	+0.13
0.58	+0.26
1.0	+0.49

Prices barely move — large-home prices rise about 1%.

- One caveat: a small conditional grant can pull purchases back before the birth, delaying them through anticipation.

Grants move the birth margin because they relax the owner financing gap $\zeta_i^{O,F}$.

A Package That Decentralizes the Reallocation

Tax the incumbent stock and spend the revenue on young buyers' access:

$$\text{large-home tax} + \text{birth-linked purchase grant} \quad \Rightarrow \quad \Delta\mathrm{TFR} = +0.15, \quad \Delta p_{\text{large}} = -11\%.$$

The tax lowers what incumbents will pay to keep large units; the grant closes the young side's financing gap at the birth margin. Revenue-funded, it is the decentralized version of the planner's move — and it works through access, not price.

The Bridge, in Three Numbers

1. Access: fertile renters at the cap have $\zeta/q = 1.28$, and 0.734 of the fertility-response mass sits on households with $\zeta > 0$.
2. Allocation: 87% of the family-sized stock is held by post-fertile households.
3. Policy: the tax-funded access transfer raises TFR by 0.15 while cutting large-home prices 11%.

The same object ζ measures fertility exposure and stock misallocation, and the package is the decentralized version of the planner's reallocation.

Where This Version Falls Short

- Several parameters sit at their search bounds.[3]
- Fertility is mistimed: births cluster at 33, against a data peak near 27.
- Planned extension: a child time cost, so children are priced partly in foregone income.
- Planned extensions: an old-age exit margin, and childlessness targets consistent with the estimand.

These belong in the draft as honest limitations, not as settled calibration details.

[3] Eight of the 13 estimated parameters are at or against a bound at the current calibration — evidence that the parameter space is compensating for missing mechanisms, not a failure of the search.

Conclusion

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